

Forces & Equilibrium



Force as a vector quantity

- 1 A particle has two forces acting on it: $\mathbf{F}_1 = (3, 4)$ N and $\mathbf{F}_2 = (-1, 2)$ N. Find the resultant force and its magnitude.
 - 2 A particle has two forces acting on it: $\mathbf{F}_1 = (5, -2)$ N and $\mathbf{F}_2 = (1, 6)$ N. Find the resultant force and its magnitude.
 - 3 A particle has two forces acting on it: $\mathbf{F}_1 = (-4, 7)$ N and $\mathbf{F}_2 = (9, -3)$ N. Find the resultant force and its magnitude.
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Resolving forces into components

- 4 A force of 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.
 - 5 A force of 80 N acts at an angle of 50° above the horizontal. Find its horizontal and vertical components.
 - 6 A force of 100 N acts at an angle of 60° above the horizontal. Find its horizontal and vertical components.
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Equilibrium of a particle under coplanar forces

- 7 A particle is in equilibrium under three forces: $(6, 0)$ N, $(-2, 5)$ N, and $\mathbf{F}$. Find $\mathbf{F}$.
 - 8 A particle of weight 20 N hangs in equilibrium, supported by two strings making angles of 40° and 60° with the ceiling on either side. Set up the two equilibrium equations (horizontal and vertical) using tensions T_1 and T_2 .
 - 9 A particle is in equilibrium under three forces: $(3, -4)$ N, $(5, 1)$ N, and $\mathbf{F}$. Find $\mathbf{F}$.
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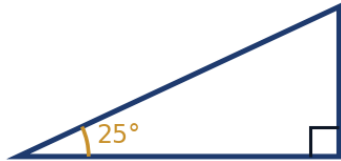
Friction and the coefficient of friction

- 10 A block of mass 5 kg rests on a rough horizontal surface with coefficient of friction $\mu = 0.3$. Find the maximum friction force before the block slides, using $F = \mu R$ and $g = 9.8 \text{ m/s}^2$.
- 11 A block of mass 8 kg rests on a rough horizontal surface. It is on the point of sliding when a horizontal force of 23.52 N is applied. Find the coefficient of friction μ , using $g = 9.8 \text{ m/s}^2$.

- 12 A block of mass 10 kg rests on a rough horizontal surface with coefficient of friction $\mu = 0.4$. Find the maximum friction force before the block slides, using $F = \mu R$ and $g = 9.8 \text{ m/s}^2$.
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Objects on an inclined plane

- 13 A block of mass 4 kg rests on a smooth (frictionless) plane inclined at 25° to the horizontal. Find the component of the weight acting along the plane, using $g = 9.8 \text{ m/s}^2$.



- 14 A block of mass 6 kg rests on a smooth (frictionless) plane inclined at 35° to the horizontal. Find the normal reaction force on the block, using $g = 9.8 \text{ m/s}^2$.
- 15 A block of mass 8 kg rests on a rough plane inclined at 30° to the horizontal. Find the component of the weight acting perpendicular to the plane, using $g = 9.8 \text{ m/s}^2$.
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Connected particles and strings over a pulley

- 16 Two masses, 6 kg and 4 kg, are connected by a light inextensible string over a smooth pulley. Find the acceleration of the system and the tension in the string, using $g = 9.8 \text{ m/s}^2$.
- 17 Two masses, 5 kg and 3 kg, are connected by a light inextensible string over a smooth pulley. Find the acceleration of the system and the tension in the string, using $g = 9.8 \text{ m/s}^2$.
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Newton's third law and normal reaction

- 18 A box of mass 8 kg rests on a horizontal table. A person pushes down on the box with a vertical force of 15 N. Find the normal reaction force from the table, using $g = 9.8 \text{ m/s}^2$.
- 19 A box of mass 12 kg rests on a horizontal table. A person pulls upward on the box with a vertical force of 20 N (not enough to lift it). Find the normal reaction force from the table, using $g = 9.8 \text{ m/s}^2$.
- 20 A box of mass 15 kg rests on a horizontal table. A person pushes straight down on the box with an additional vertical force of 30 N. Find the normal reaction force from the table, using $g = 9.8 \text{ m/s}^2$.

A well-designed word problem: a crate on a ramp with friction

- 21** This question has two parts. A crate of mass 10 kg sits on a ramp inclined at 20° to the horizontal, with coefficient of friction $\mu = 0.25$ between the crate and the ramp. Use $g = 9.8 \text{ m/s}^2$. (a) Find the normal reaction force on the crate. (b) Determine whether the crate remains stationary, by comparing the component of gravity along the ramp with the maximum available friction force.